AMERICAN TERAWATT

FERVO

Electrical Connections

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Insulation Coordination

Overvoltages and insulation strength

Insulation coordination

IEEE 1313

"The selection of **insulation strength** consistent with expected **overvoltages** to obtain an acceptable risk of failure."

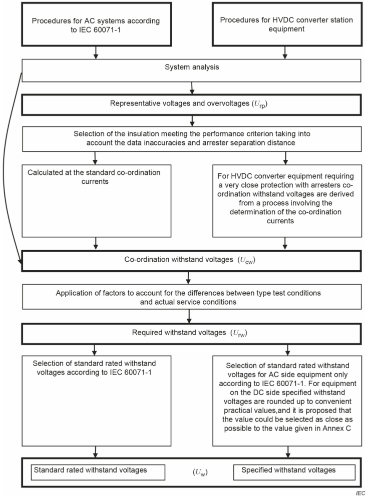
Scope of this module:

- Overvoltages
- Insulation strength
- Air clearances
- Arresters

Withstand Voltage

Withstand voltage definition process

1. Determination of the representative overvoltages U_{rp}
2. Determination of the co-ordination withstand voltages U_{cw}.
   For HVDC it can be assumed:
   $$U_{cw} = U_{rp}$$
3. Determination of the required withstand voltages U_{rw}
4. Determination of the specified withstand voltages U_w



IEC 60071-11:2022.

Representative overvoltages (U_{rp})

IEC 60071-11:2022

"... representative overvoltages are equal to protection levels of arresters for directly protected equipment."

- The statement is misleading: the arrester protection level is defined by the arrester design and the protection-coordination current amplitude and shape (lightning, switching).
- Representative overvoltages can be determined by assuming maximum system currents and switching impulse, or by transient-simulation analysis of relevant faults.
- Calculation and simulation results without any margin are considered as the representative overvoltage U_{rp}.

Required withstand voltages (U_{rw})

U_{rw} is defined as:

$$U_{rw} = K_a \cdot K_s \cdot U_{rp}$$

K_a Ambient condition correction factor

- Altitude correction
- Temperature / humidity correction for indoor installations

K_s Safety factor

- Altitude correction up to 1000m
- allowance for ageing of insulation
- allowance for changes in arrester characteristics
- allowance for dispresion in the product quality

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Required withstand voltages (U_{rw})

Table 3 – Indicative values of ratios of required impulse withstand voltage to impulse protective level

Type of equipment	Indicative values of required impulse withstand voltage/impulse protective level[a, c]		
	RSIWV/SIPL	RLIWV/LIPL	RSFIWV/STIPL[b]
AC switchyard – busbars, outdoor insulators, and other conventional equipment	1,20	1,25	1,25
AC filter components	1,15	1,25	1,25
Transformers (in oil)			
line side	1,20	1,25	1,25
valve side	1,15	1,20	1,25
Converter valves	1,15	1,15	1,20
DC valve hall equipment	1,15	1,15	1,25
DC switchyard equipment (including DC filters etc. and DC reactor)	1,15	1,20	1,25

[a] Indicated values are stated for general design objectives only. Appropriate final ratios (higher or lower) can be selected according to the chosen performance criteria.

[b] STIPL for LCC valve arresters.

[c] Indicative ratios are on the basis that any equipment is directly protected with a surge arrester.

IEC 60071-11:2022.

K_S Safety factor

Specified withstand voltages (U_w)

- Values are equal to or higher than U_{rw}.
- For AC equipment they correspond to the standard values in IEC 60071-1.
- For HVDC equipment they are rounded up to convenient values (in most cases IEC 60071-1).

Proposed U_w / kV

550	1300
650	1425
750	1550
850	1675
950	1800
1050	1950
1175	

Example for insulation point coordination

Parameter	Value
CCOV	515 kV
Number of parallel columns	8
Energy capability	17 MJ

Parameter	Value
SIPL (U_{rp} for SI)	807 kV @ 1 kA
LIPL (U_{rp} for LI)	872 kV @ 5 kA